~~CSE~~ 341 after Eid | 30/03/2026 class- Advanced Programming

Shayan

NOTE : Next বুধবার lab mid
Mid Term examinationর সিলেবাস, marks — 20/30

String handling থেকে একটা question থাকবেই

Second CT date = ??

PDF from book

- Java Strings are immutuable. <<java.lang.String>>
- String, StringBuffer, stringBuilder are FINAL
  None of them can be subclassed.
  - These three implement CharSequence interface.
- A string object cannot be changed after creation.

```
psvm(...){
    String s = "ABC";
    s.concat("DEF");
    System.out.println(s);  // prints only "ABC"
}
```

```
psvm(...){
    String s = "ABC";
    s = s.concat("DEF");    // assignment operation
    System.out.println(s);  // ABCDEF
}
```

$\boxed{\text{"+" operator = concat()}}$ // concatination



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## String constructors

Default = new String()

Parameterized =

① Array char chars[] = {"a", "b", "c"};

String s = new String(chars);

② Range fixing

String(char chars[], int startIndex, int numChars)

```
psvm(...) {
    char chars[] = {'a', 'b', 'c', 'd', 'e', 'f'};
    String s = new String(chars, 2, 3);
    System.out.println(s);   // cde
```

## String length

$\boxed{\text{s.length();}}$

empty string length = 0

(!) empty string ≠ NULL

## String Operations

① Concatenation

{LTR operation}

i) s = "four" + 2 + 2 ⇒ four22

ii) s = "four" + (2+2) ⇒ four4

② toString

⇒ Return Type always String হবে

i) public keyword use করতে হবে (during method overriding)

ii) calls automatically

ii) Java built-in method

When toString() is called automatically? Shayan

→ during ~~System.out.~~ System.out.println("")

→ during concatenation of string & objects.

```
class Box{
    int w,h;
    Box(int w, int h){
        this.w= w;
        this.h=h;
    }
    public String toString(){
        return "Dimension are: "+ w+ " by "+ h +".";
    }
}
public class Main{
    public static void main(String[] args){
        Box b= new Box(10,5);
        String s = "Box "+b; // String s = b
        System.out.println(b);
        System.out.println(s);
    }
}
```

output?

"toString() can be overriden in any class"



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## String operations

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### ③ Character Extractions

char charAt(int where)  single character extraction

```
char s= "abc".charAt(1);
System.out.println(s);   // b
```

multi character extraction

```
void getChars(int sourceStart,
              int sourceEnd,
              char target[],
              int targetStart)
```

```
psvm(...){
    String s= "This is a demo of the get chars Method.";
    int start =10;
    int end =14;
    char buf[]= new char[end-start];
    s.getChars(start, end, buf,0);
    System.out.println(buf);
```

### ④ String comparison (boolean type)

→ boolean equals(Object str)  //case sensitive

→ boolean equalsIgnoreCase(String str)  //case insensitive

# self study : $\boxed{\text{equals}}$ vs $\boxed{==}$

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# String Operations

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## (5) String Searching

→ indexOf(String str, int startIndex) → 0 10 20

→ lastIndexOf(String str, int startIndex) 0 10 20

indexOf(t) = 7

LastIndexOf(t) = ~~54 64~~ 65

0 3 6 7 10 11 15 19 23 28 32

Now is the time for all good men

35 40 43 47 51 54 55 60

to come to the aid of their ~~company~~ 65 67

country

indexOf(the) = 7

lastIndexOf(the) = 55

indexOf(t, 10) = 11

lastIndexOf(t, 60) = 7

indexOf(the, 10) =

lastIndexOf(the, 60) = 55

## (6) Modifying a String

→ substring(int startIndex, int endIndex)

→ concat

→ replace(char original, char replacement) //all occurances

→ trim // remove whitespaces

~~→ chang~~



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## String operations

⑦ changing cases

① toLowerCase();
② toUpperCase();



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